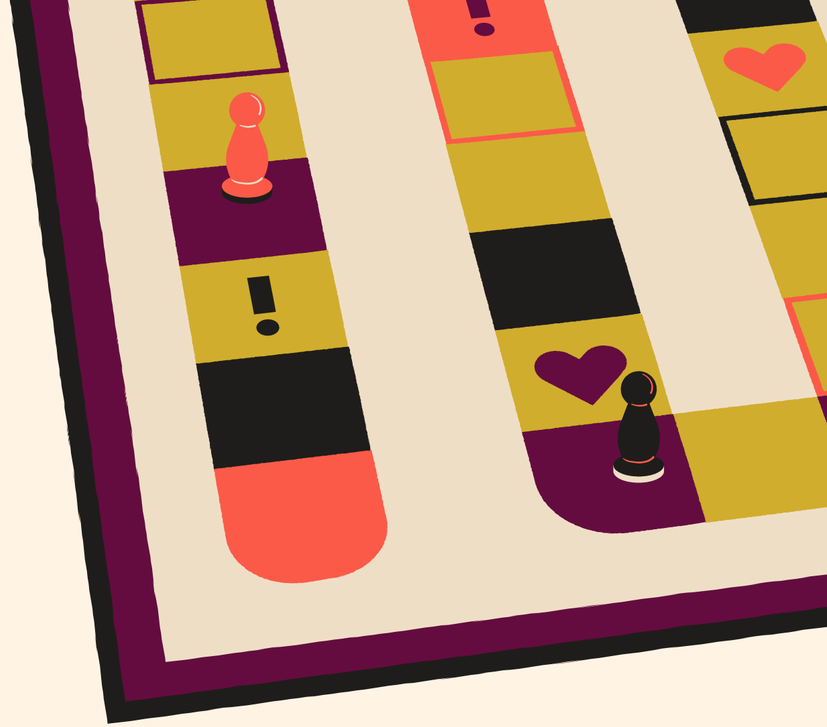
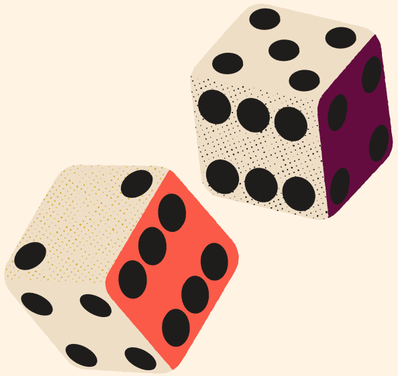
BOARD GAME



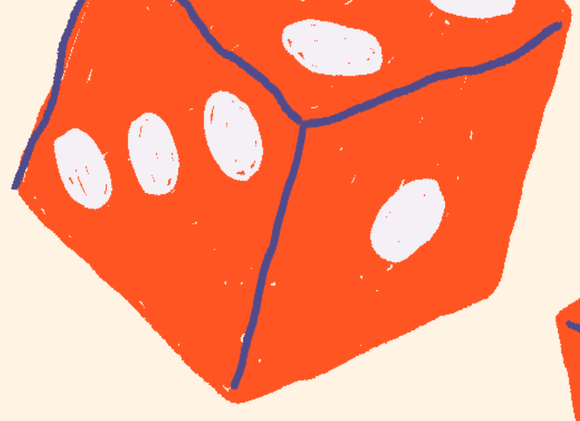
DATA SET



Kristina Đurić



HOW TO PLAY





- Skup podataka sadrži podatke o društvenim igrama poput imena, kategorije, min-max igrača, izdavača, itd od 80tih do 2019.
- Početni skup podataka sadrži 81 kolonu i 90.400 redova
- Veličina skupa podataka: 397Mb






LET'S GET STARTED!

INICIJALNA LOGIČKA ŠEMA

BoardGames		CREATE TABLE "BoardGames" ("row
row_names	TEXT	"row_names" TEXT
game.id	TEXT	"game.id" TEXT
game.type	TEXT	"game.type" TEXT
details.description	TEXT	"details.description" TEXT
details.image	TEXT	"details.image" TEXT
details.maxplayers	INTEGER	"details.maxplayers" INTEGER
details.maxplaytime	INTEGER	"details.maxplaytime" INTEGER
details.minage	INTEGER	"details.minage" INTEGER
details.minplayers	INTEGER	"details.minplayers" INTEGER
details.minplaytime	INTEGER	"details.minplaytime" INTEGER
details.name	TEXT	"details.name" TEXT
details.playingtime	INTEGER	"details.playingtime" INTEGER
details.thumbnail	TEXT	"details.thumbnail" TEXT
details.yearpublished	INTEGER	"details.yearpublished" INTEGER
attributes.boardgameartist	TEXT	"attributes.boardgameartist" TEXT
attributes.boardgamecategory	TEXT	"attributes.boardgamecategory" TEXT
attributes.boardgamecompilation	TEXT	"attributes.boardgamecompilation" TEXT
attributes.boardgamedesigner	TEXT	"attributes.boardgamedesigner" TEXT
attributes.boardgameexpansion	TEXT	"attributes.boardgameexpansion" TEXT
attributes.boardgamefamily	TEXT	"attributes.boardgamefamily" TEXT
attributes.boardgameintegration	TEXT	"attributes.boardgameintegration" TEXT
attributes.boardgamemechanic	TEXT	"attributes.boardgamemechanic" TEXT
attributes.boardgamepublisher	TEXT	"attributes.boardgamepublisher" TEXT
attributes.total	REAL	"attributes.total" REAL
stats.average	REAL	"stats.average" REAL
stats.averageweight	REAL	"stats.averageweight" REAL
stats.bayesaverage	REAL	"stats.bayesaverage" REAL
stats.family.abstracts.bayesaverage	REAL	"stats.family.abstracts.bayesaverage" REAL
stats.family.abstracts.pos	REAL	"stats.family.abstracts.pos" REAL
stats.family.cgs.bayesaverage	REAL	"stats.family.cgs.bayesaverage" REAL
stats.family.cgs.pos	REAL	"stats.family.cgs.pos" REAL
stats.family.childrensgames.bayesa...	REAL	"stats.family.childrensgames.bayesaverage" REAL
stats.family.childrensgames.pos	REAL	"stats.family.childrensgames.pos" REAL
stats.family.familygames.bayesaver...	REAL	"stats.family.familygames.bayesaverage" REAL
stats.family.familygames.pos	REAL	"stats.family.familygames.pos" REAL
stats.family.partygames.bayesavera...	REAL	"stats.family.partygames.bayesaverage" REAL
stats.family.partygames.pos	REAL	"stats.family.partygames.pos" REAL
stats.family.strategygames.bayesav...	REAL	"stats.family.strategygames.bayesaverage" REAL
stats.family.strategygames.pos	REAL	"stats.family.strategygames.pos" REAL
stats.family.thematic.bayesaverage	REAL	"stats.family.thematic.bayesaverage" REAL
stats.family.thematic.pos	REAL	"stats.family.thematic.pos" REAL
stats.family.wargames.bayesaverage	REAL	"stats.family.wargames.bayesaverage" REAL











INICIJALNA LOGIČKA ŠEMA



polls.suggested_numplayers.Over	TEXT	"polls.suggested_numplayers.Over" TEXT
polls.suggested_playerage	TEXT	"polls.suggested_playerage" TEXT
attributes.t.links.concat.2....	TEXT	"attributes.t.links.concat.2...." TEXT
stats.family.amiga.bayesaverage	REAL	"stats.family.amiga.bayesaverage" REAL
stats.family.amiga.pos	REAL	"stats.family.amiga.pos" REAL
stats.family.arcade.bayesaverage	REAL	"stats.family.arcade.bayesaverage" REAL
stats.family.arcade.pos	REAL	"stats.family.arcade.pos" REAL
stats.family.atarist.bayesaverage	REAL	"stats.family.atarist.bayesaverage" REAL
stats.family.atarist.pos	REAL	"stats.family.atarist.pos" REAL
stats.family.commodore64.bayesav...	REAL	"stats.family.commodore64.bayesaverage" REAL
stats.family.commodore64.pos	REAL	"stats.family.commodore64.pos" REAL
stats.subtype.rpgitem.bayesaverage	REAL	"stats.subtype.rpgitem.bayesaverage" REAL
stats.subtype.rpgitem.pos	REAL	"stats.subtype.rpgitem.pos" REAL
stats.subtype.videogame.bayesaver...	REAL	"stats.subtype.videogame.bayesaverage" REAL
stats.subtype.videogame.pos	REAL	"stats.subtype.videogame.pos" REAL

stats.median	REAL	"stats.median" REAL
stats.numcomments	REAL	"stats.numcomments" REAL
stats.numweights	REAL	"stats.numweights" REAL
stats.owned	REAL	"stats.owned" REAL
stats.stddev	REAL	"stats.stddev" REAL
stats.subtype.boardgame.bayesaver...	REAL	"stats.subtype.boardgame.bayesaverage" REAL
stats.subtype.boardgame.pos	REAL	"stats.subtype.boardgame.pos" REAL
stats.trading	REAL	"stats.trading" REAL
stats.usersrated	REAL	"stats.usersrated" REAL
stats.wanting	REAL	"stats.wanting" REAL
stats.wishing	REAL	"stats.wishing" REAL
polls.language_dependence	TEXT	"polls.language_dependence" TEXT
polls.suggested_numplayers.1	TEXT	"polls.suggested_numplayers.1" TEXT
polls.suggested_numplayers.10	TEXT	"polls.suggested_numplayers.10" TEXT
polls.suggested_numplayers.2	TEXT	"polls.suggested_numplayers.2" TEXT
polls.suggested_numplayers.3	TEXT	"polls.suggested_numplayers.3" TEXT
polls.suggested_numplayers.4	TEXT	"polls.suggested_numplayers.4" TEXT
polls.suggested_numplayers.5	TEXT	"polls.suggested_numplayers.5" TEXT
polls.suggested_numplayers.6	TEXT	"polls.suggested_numplayers.6" TEXT
polls.suggested_numplayers.7	TEXT	"polls.suggested_numplayers.7" TEXT
polls.suggested_numplayers.8	TEXT	"polls.suggested_numplayers.8" TEXT
polls.suggested_numplayers.9	TEXT	"polls.suggested_numplayers.9" TEXT









ČIŠĆENJE PODATAKA





- Drop svih kolona koje iamju više od 60% null vrednosti
- Preimenovanje kolona
- Postavljanje O vrednosti na null vrednosti u nekim kolonama
- Brisanje nepotpunih redova
- Broj redova i kolona nakon sređivanja: (88966, 29)
- Veličina skupa podataka nakon čišćenja: 146 Mb





SEMANTIKA OBELEŽJA

- **id –** jedinstveni ID igre u bazi.boardgamecategory
- **category** – spisak kategorija igre (npr. „Strategy“, „Party“, „Family“).
- **categories**
- **name –** naziv igre
- **boardgamedesigner –** imena dizajnera igre.
- **boardgamefamily –** familija / serija kojoj igra pripada (npr. „Catan Series“).
- **boardgamemechanic –** mehanike igranja (npr. „Deck Building“, „Worker Placement“).
- **boardgamepublisher –** izdavači igre.
- **description –** tekstualni opis igre.
- **maxplayers –** maksimalan broj igrača.
- **maxplaytime –** maksimalno trajanje partije u minutima.
- **minage –** minimalna preporučena starost igrača.
- **minplayers –** minimalan broj igrača.
- **minplaytime –** minimalno trajanje partije u minutima.
- **playingtime –** prosečno / tipično trajanje partije.



SEMANTIKA OBELEŽJA

- **playingtime –** prosečno / tipično trajanje partije.
- **type –** tip entiteta (npr. „boardgame“, „expansion“).
- **average –** prosečna ocena igre (od korisnika).
- **yearpublished –** godina kada je igra prvi put objavljena.
- **averageweight –** prosečna „težina“ (koliko je kompleksna igra).
- **numcomments –** broj komentara korisnika.
- **numweights –** broj korisnika koji su ocenili „težinu“.
- **owned –** koliko ljudi je prijavilo da poseduje igru.
- **trading –** koliko ljudi nudi igru na zamenu.
- **usersrated –** broj ljudi koji su dali ocenu.
- **wanting –** koliko ljudi je reklo da želi igru.
- **wishing –** koliko ljudi je stavilo igru na listu želja.
-



SPELL IT CORRECTLY!

Level 1




```
{
    "_id" : ObjectId("68dc38a10652bd64d867b46c"),
    "category" : "Exploration,Horror",
    "designer" : "Friedemann Friese",
    "mechanic" : "Modular Board",
    "publisher" : "2F-Spiele",
    "description" : "Each player in Frischfleish (Fresh Flesh) represents a unique group of individuals stranded on a desert island.  Although there is a rescue attempt in the works, it's still a month away.  In the meantime, the characters will have to live off the land or off of each other depending on how it goes.  Players attempt to guide the members of their group around the island, keeping them alive with the food they find in the desert.  However if their foraging is less than successful, finding some poor schmuck's meaty thigh is a viable substitute.&#10;&#10;",
    "maxplayers" : NumberInt(6),
    "maxplaytime" : NumberInt(120),
    "minage" : NumberInt(18),
    "minplayers" : NumberInt(2),
    "minplaytime" : NumberInt(120),
    "name" : "Frischfleisch",
    "playingtime" : NumberInt(120),
    "yearpublished" : NumberInt(1999),
    "game_id" : "133",
    "game_type" : "boardgame",
    "row_names" : "125",
    "average" : 6.12852,
    "averageweight" : 2.5455,
    "bayesaverage" : 5.58356,
    "median" : NumberInt(0),
    "numcomments" : NumberInt(57),
    "numweights" : NumberInt(11),
    "owned" : NumberInt(211),
    "stddev" : 1.38487,
    "trading" : NumberInt(6),
    "usersrated" : NumberInt(142),
    "wanting" : NumberInt(59),
    "wishing" : NumberInt(100),
    "categories" : [
        "Exploration",
        "Horror"
    ]
}
```

CHOOSE YOUR PAWN

Upiti 1, 2, 3, 4 i 5



Red pawn
Upit 1



Yellow pawn
Upit 4



Blue pawn
Upit 2



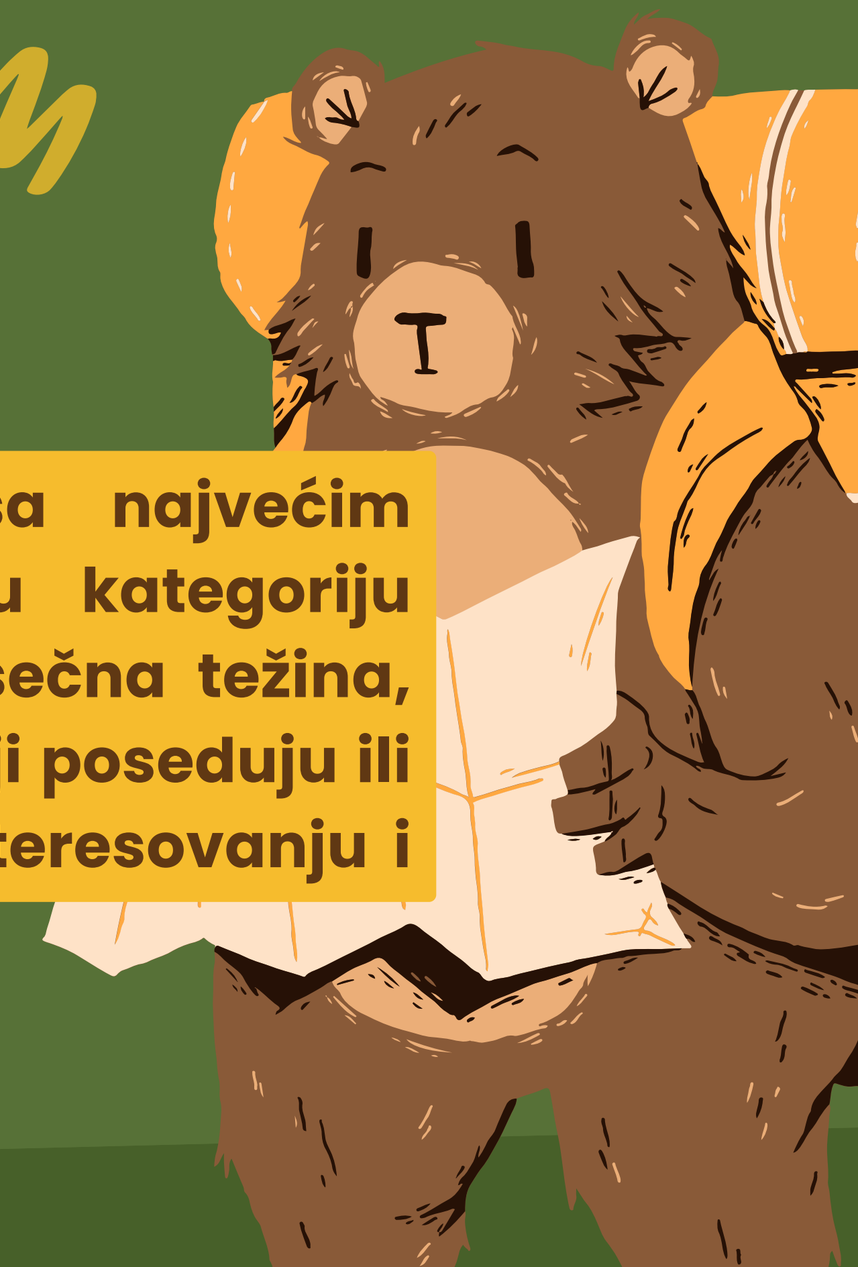
Pink pawn
Upit 5



White pawn
Upit 3

Answer Key

DID YOU GET THE RIGHT ANSWER?



Prikazati 10 kategorija društvenih igara sa najvećim interesovanjem korisnika, gde se za svaku kategoriju izračunavaju broj igara, prosečna ocena, prosečna težina, prosečan broj igrača i ukupan broj korisnika koji poseduju ili žele igru. Rezultati su sortirani po ukupnom interesovanju i prosečnoj oceni.



```
db.BoardGames.aggregate([
  { $match: { category: { $exists: true, $ne: null }, average: { $gt: 0 } } },
  { $addFields: {
      wants_game: { $add: [{ $getField: { field: "wanting", input: "$$ROOT" } }, { $getField: { field: "wishing", input: "$$ROOT" } },
          { $getField: { field: "trading", input: "$$ROOT" } }] },
      owned_plus_wants: { $add: [{ $getField: { field: "owned", input: "$$ROOT" } }, { $add: [{ $getField: { field: "wanting", input: "$$ROOT" } },
           { $getField: { field: "wishing", input: "$$ROOT" } }, { $getField: { field: "trading", input: "$$ROOT" } }] }] },
      avg_players: { $avg: [{ $getField: { field: "minplayers", input: "$$ROOT" } }, { $getField: { field: "maxplayers", input: "$$ROOT" } }] },
      category: { $getField: { field: "category", input: "$$ROOT" } },
      avg_rating: { $getField: { field: "average", input: "$$ROOT" } },
      avg_weight: { $getField: { field: "averageweight", input: "$$ROOT" } }
    }
  },
  { $group: { _id: "$category", game_count: { $sum: 1 }, avg_rating: { $avg: "$avg_rating" }, avg_weight: { $avg: "$avg_weight" },
      total_owned_plus_wants: { $sum: "$owned_plus_wants" }, avg_players: { $avg: "$avg_players" } } },
  { $sort: { total_owned_plus_wants: -1, avg_rating: -1 } },
  { $project: { _id: 0, category: "$_id", game_count: 1, avg_rating: { $round: ["$avg_rating", 2] }, avg_weight: { $round: ["$avg_weight", 2] },
      avg_players: { $round: ["$avg_players", 1] }, total_owned_plus_wants: 1 } },
  { $limit: 10 }
])
```

Answer Key

UPIT 2

Prikazati sve igre koje imaju više od 500 ocena, prosečnu ocenu veću od 6.5, težinu igre veću od 4 i definisanog izdavača. Za svaku igru izračunati prosečan broj igrača a zatim grupisati igre po izdavaču kako bi se prikazale sledeće informacije: ukupni broj izdatih igara, zbir korisničkih ocena, prosečnu težinu igara, prosečan broj igrača, kao i najbolju igru po oceni (naziv i tip igre). Prikazati top 5 izdavača po ukupnom broju izdatih igara.



```
db.BoardGamesOptimized.aggregate([
  { $match: { "Stats.UsersRated": { $gt: 500 }, "Stats.Average": { $gt: 6.5 }, "Stats.AverageWeight": { $gt: 4 }, "Design.Publisher": { $exists: true, $ne: null
  { $addFields: { avg_players: { $avg: [{ $toDouble: "$Players.MinPlayers" }, { $toDouble: "$Players.MaxPlayers" }] } } },
  { $group: {
      _id: "$Design.Publisher",
      total_games: { $sum: 1 },
      total_usersrated: { $sum: "$Stats.UsersRated" },
      avg_weight: { $avg: "$Stats.AverageWeight" },
      avg_players: { $avg: "$avg_players" },
      best_game: { $first: "$$ROOT" }
    }
  },
  { $sort: { total_games: -1 } },
  { $limit: 5 },
  { $project: {
      _id: 0,
      publisher: "$_id",
      total_games: 1,
      total_usersrated: 1,
      avg_weight: { $round: ["$avg_weight", 2] },
      avg_players: { $round: ["$avg_players", 1] },
      best_game_name: "$best_game.Name",
      best_game_type: "$best_game.Type",
      best_game_rating: "$best_game.Stats.Average"
    }
```

Answer Key

UPIT 3

Prikazati top 3 izdavača igara čije prosečne ocene prelaze prosečnu ocenu svih igara. Za svakog izdavača prikazati broj igara, zbir korisničkih ocena i prosečnu težinu igara, kao i sumirani pregled top 3 izdavača sa ovim podacima.



```
db.BoardGames.aggregate([
  { $match: { "Stats.Average": { $gt: 0 } } },
  { $group: { _id: null, avg_all_games: { $avg: "$Stats.Average" }, all_games: { $push: "$$ROOT" } } },
  { $project: { high_rating: { $filter: { input: "$all_games", as: "g", cond: { $gte: ["$$g.Stats.Average", "$avg_all_games"] } } } } },
  { $unwind: "$high_rating" },
  { $project: { publishers_array: { $split: [{ $ifNull: ["$high_rating.Design.Publisher","Unknown"] }, ","] }, usersrated: "$high_rating.Stats.UsersRate
      avgweight: "$high_rating.Stats.AverageWeight" } },
  { $unwind: "$publishers_array" },
  { $group: { _id: { $trim: { input: "$publishers_array" } }, total_games: { $sum: 1 }, total_usersrated: { $sum: "$usersrated" },
      avg_weight: { $avg: "$avgweight" } } },
  { $project: { total_games: 1, total_usersrated: 1, avg_weight: { $round: ["$avg_weight",2] } } },
  { $sort: { total_games: -1 } },
  { $limit: 3 },
  { $group: { _id: "high_rating_summary", top_publishers: { $push: "$_id" }, total_games: { $sum: "$total_games" },
      total_usersrated: { $sum: "$total_usersrated" }, avg_weight: { $avg: "$avg_weight" } } },
  { $project: { top_publishers: 1, total_games: 1, total_usersrated: 1, avg_weight: { $round: ["$avg_weight",2] } } }
])
```

Answer Key

UPIT 4

Prikazati sve igre objavljene od 2010. godine koje spadaju u kategorije “Strategy” ili “Card Game”, imaju najmanje 50 ocena i maksimalno 4 igrača. Za svaku igru prikazati naziv, izdavača, godinu izdavanja, prosečnu ocenu i opseg broja igrača u formatu "min-max igrača", sortirano po prosečnoj oceni silazno.



```
db.BoardGames.aggregate([
  { $match: { YearPublished: { $gte: 2010 }, "Design.Categories": { $in: ["Strategy","Card Game"] },
      "Stats.UsersRated": { $gte: 50 }, "Players.MaxPlayers": { $lte: 4 } } },
  { $addFields: { players_range: { $concat: [{ $toString: "$Players.MinPlayers" }, "-",
      { $toString: "$Players.MaxPlayers" }, " igrača"] } } },
  { $sort: { "Stats.Average": -1 } },
  { $project: { Name: 1, Publisher: 1, YearPublished: 1, "Stats.Average": 1, players_range: 1 } }
])
```

Answer Key

UPIT 5

Prikazati 10 najboljih strategijskih igara sa vremenom igranja manjim od 60 minuta, više od 500 ocena i prosečnom ocenom većom od 7. Za svaku igru prikazati naziv, vreme igranja, minimalni i maksimalni broj igrača, sortirano po prosečnoj oceni silazno.



```
db.BoardGames.aggregate([
  { $match: { "Design.Categories": /Strategy/, "Players.PlayingTime": { $lt: 60 },
      "Stats.UsersRated": { $gte: 500 }, "Stats.Average": { $gt: 7 } } },
  { $sort: { "Stats.Average": -1 } },
  { $limit: 10 },
  { $project: { _id: 0, Name: 1, "Players.PlayingTime": 1, "Players.MinPlayers": 1,
      "Players.MaxPlayers": 1 } }
])
```

LOGIČKA ŠEMA ZA OPTIMIZACIJU

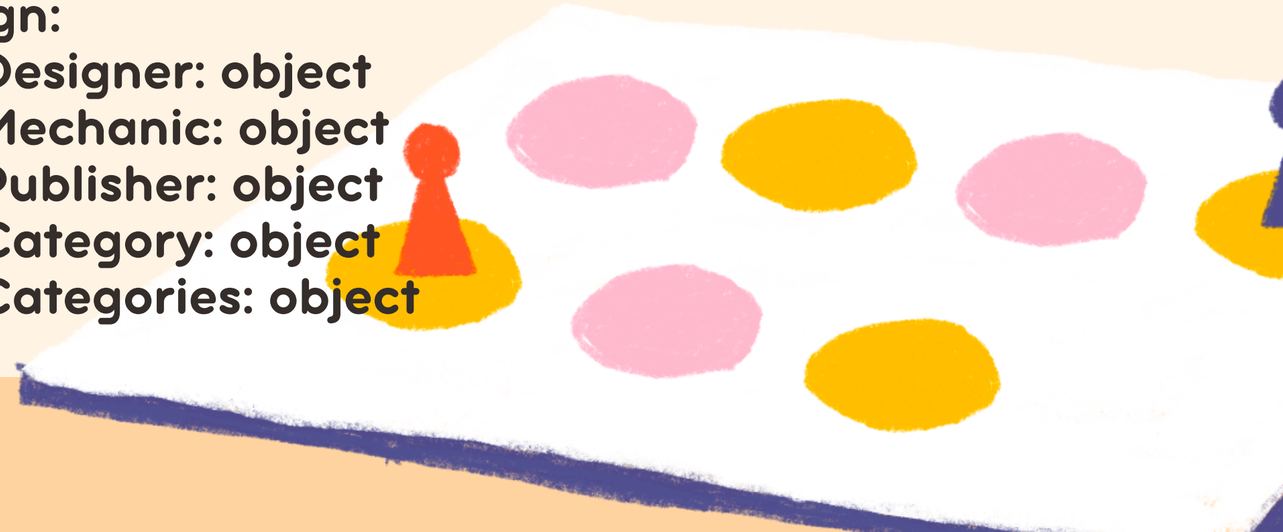
Game
- GameId: object
- Name: object
- Description: object
- YearPublished: float
- Type: object (expansion, game)

Stats:
- Average: float
- Median: float
- AverageWeight: float
- NumComments: float
- NumWeights: float
- UsersRated: float
- Owned: float
- Trading: float
- Wanting: float
- Wishing: float

Players
- MinPlayers: object
- MaxPlayers: object
- MinPlayTime: float
- MaxPlayTime: float
- PlayingTime: float
- MinAge: float

Design:
- Designer: object
- Mechanic: object
- Publisher: object
- Category: object
- Categories: object





Answer Key

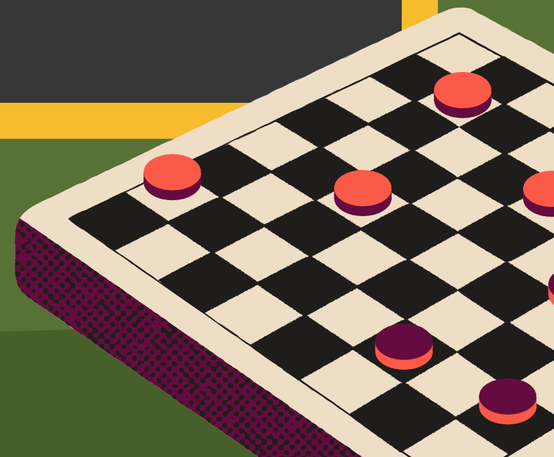
BUCKETING

```python
def process_and_write_bucket(bucket, output_file):
    restructured_games = []
    for game in bucket:
        restructured_game = process_game(game)
        restructured_games.append(restructured_game)

    with open(output_file, 'a', encoding='utf-8') as file:
        for game in restructured_games:
            json.dump(game, file)
            file.write('\n')

def read_in_buckets(file_path, bucket_size):
    with open(file_path, 'r', encoding='utf-8') as file:
        bucket = []
        for line in file:
            game = json.loads(line)
            bucket.append(game)
            if len(bucket) >= bucket_size:
                yield bucket
                bucket = []
        if bucket:
            yield bucket
```







CONGRATULATIONS WINNER!

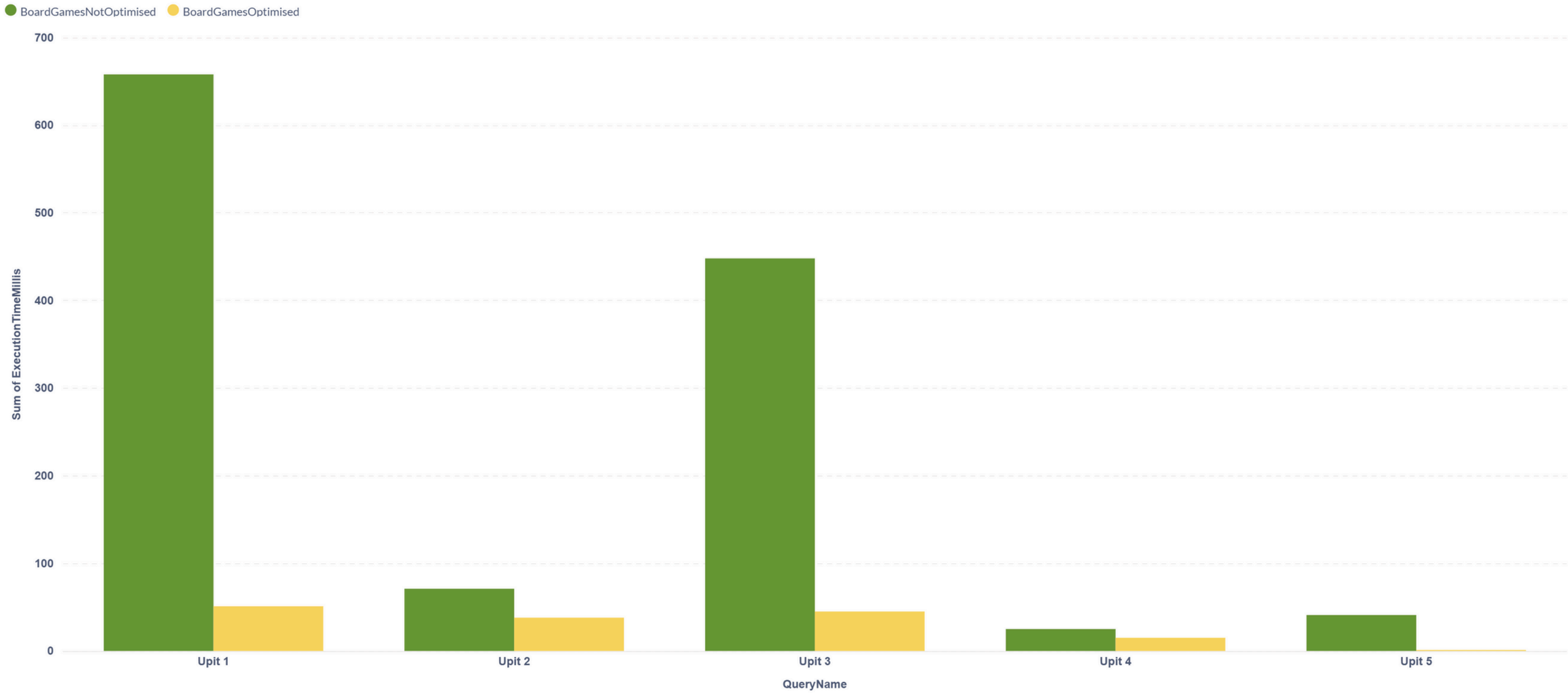



```
db.BoardGamesOptimized.createIndex({"Design.Category": 1, "Stats.Average": -1 })
db.BoardGamesOptimized.createIndex({"Stats.UsersRated": 1, "Stats.Average": 1, "Stats.AverageWeight": 1, "Design.Publisher": 1})
db.BoardGamesOptimized.createIndex({"Stats.Average": -1, "Stats.UsersRated": 1, "Design.Categories": 1, "Players.MaxPlayers": 1})
db.BoardGamesOptimized.createIndex({"YearPublished": 1, "Design.Categories": 1, "Stats.UsersRated": 1, "Players.MaxPlayers": 1, "Stats.Average": -1
```

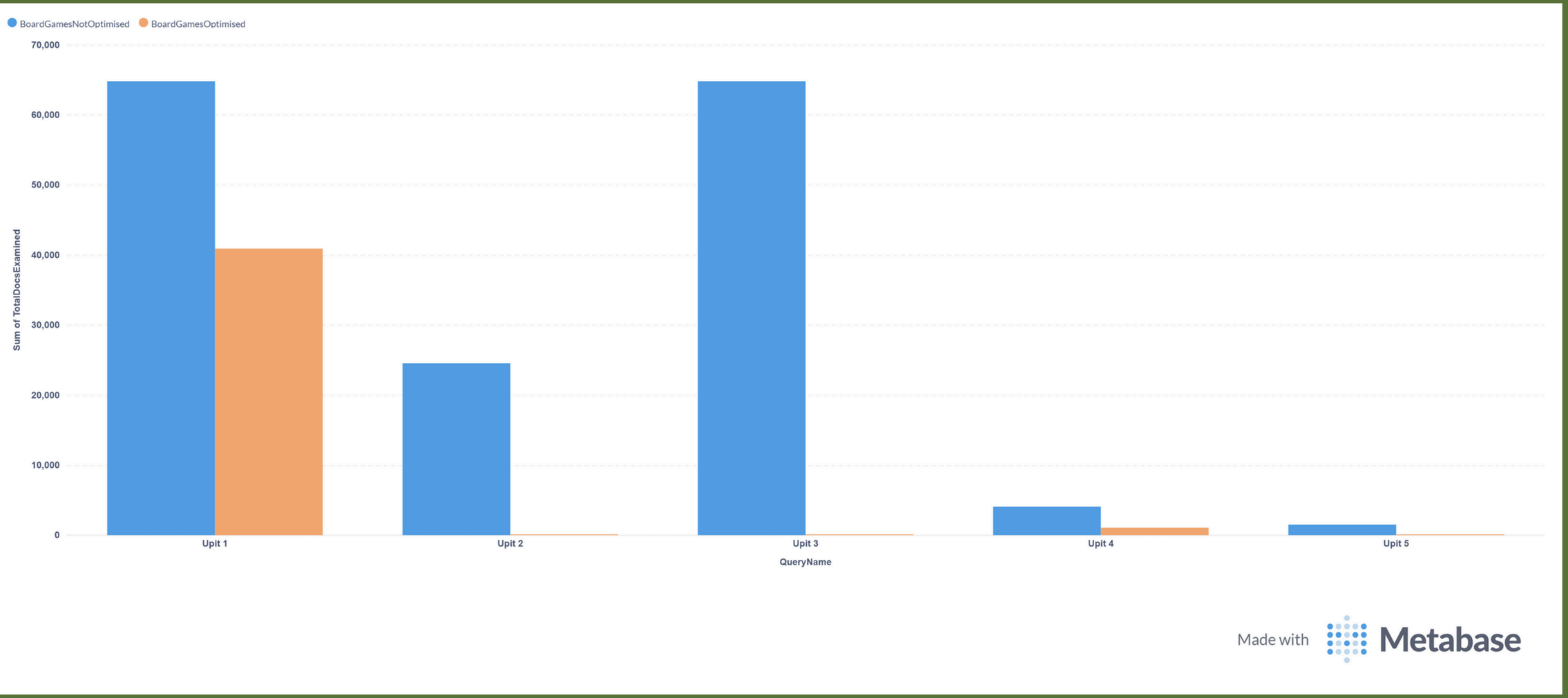








Made with Metabase


BoardGamesNotOptimised
BoardGamesOptimised
70,000
60,000
50,000
40,000
30,000
20,000
10,000
0
Sum of TotalDocsExamined
Upit 1
Upit 2
Upit 3
Upit 4
Upit 5
QueryName
Made with Metabase

See you at the next game!

THANKS FOR PLAYING



